

Can one reasoning agent turn heterogeneous site evidence into a persistent, editable digital twin?

Physical evidence goes in; the same GPT-6 Astra/Codex engineering agent repeatedly proposes programs and revisions; validated component models accumulate into one station state.

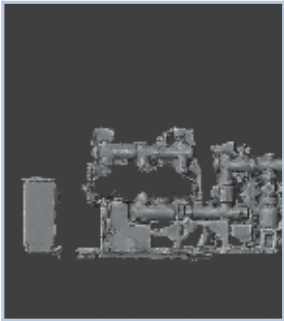
1 · EVIDENCE FROM THE OPERATING SITE

Field photographs



visible parts, counts, orientation and appearance

Registered coarse geometry



metric layout, occupied surfaces and omissions

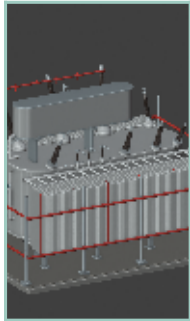
2 · GPT-6 ASTRA + CODEX

3D engineering agent

observe evidence
→ form / revise a geometric hypothesis
→ write or modify Python / Blender actions
→ read review + validator feedback
same agent · many tasks · persistent state

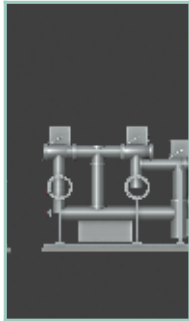
3 · EDITABLE TASK OUTPUTS

Transformer



shared MASTER + site instance

GIS



component family + site geometry

ENGINEERING CONTEXT + HISTORY

inventory / 12MZ semantics · prior Blender state · validators
human review · failed revisions · manifests / continuation notes

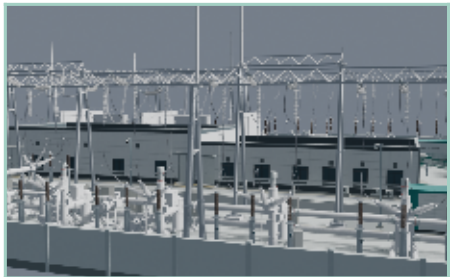
OTHER TASKS

arresters · capacitor banks · conductors · buildings
auxiliary facilities · human-guided correction · semantics

WHAT MAKES THIS AN AGENT STUDY?

The artifact persists across tasks. Later batches inherit earlier geometry, assets, validators and mistakes. Failures can change what is measured, what is shared, and what is built next.

4 · Persistent station-scale digital twin



37,153 scene objects · shared assets · connections · history · later inspection semantics

Figure 1 · Study overview. The photogrammetric reconstruction is evidence, not the final representation. AstraBuild targets an editable, componentized and traceable engineering state that accumulates across heterogeneous reconstruction tasks.